

Notation Guide

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The Mathematics of Artificial Intelligence Agents

A Readable Guide to Decisions, Planning, Memory, Tools, Learning, and Cooperation

Chapter links in this companion copy open the corresponding laboratory reading page. The symbols and explanations are retained from the book; its original notation source is preserved in [reference-sources/notation-guide.md](#).

Core symbols carry their meanings across chapters. Explicitly declared local aliases and diagram labels are scoped where they appear; a local count or vertex label does not redefine a global quantity.

That rule costs something. It means a few symbols are less conventional than the ones a specialist in one subfield would expect, because a letter that is standard in information theory may already be doing a job in decision theory. Where this book departs from a field's usual notation, it is to avoid a collision, and the departure is noted at the point of use.

This guide is for returning to, not for memorizing. Every symbol is also defined where it first appears.

Conventions

- Scalars are italic; vectors are bold lowercase; matrices are uppercase.
- Sets, spaces, and operators use calligraphic or roman forms.
- Random variables are uppercase and their realized values are the matching lowercase letter, so the composite state random variable is written one way and a particular value of it another.
- Named operations such as `dec`, `Mem`, and `Reach` are roman, never italic single letters.
- Probability is $\Pr$ and expectation is $\mathbb{E}$. Chapters 1 and 22 typeset the expectation operator as $\mathbb{E}$; Chapters 6, 7, 10, 11, 13, 15, 16, and 19 typeset the same operator as roman E (E or E). Both forms denote one operator. A plain E used as a subscript label, such as a delegated destination in Chapter 27, names a thing and is not an operator.
- Every logarithm states its base or its units at first use.
- Subscript t is time, except where a chapter declares otherwise (Chapter 19's matrix M_{st} uses s and t as run indices). Subscript m indexes a member of a declared model family.
- A few letters change meaning between chapters, for example Γ , α , δ , τ , C , z , u , Φ , and λ . Each meaning is scoped to the chapter listed beside it and restated where it applies.

The two quantities the book is built on

Two objects recur from Part I onward and are worth reading before Chapter 1.

The score of an assembly. $U_{\mathcal{V}}(S) = \mathbb{E}[Y_S \mid \mathcal{V}]$ is the population score of the system built from enabled component set S , under declared evaluation contract $\mathcal{V}$; Y_S is the score of one run of that system. $\hat{U}_{\mathcal{V}}(S)$ is its finite-sample estimate. The contract fixes task distribution, scoring rule and units, sampling settings, resource budget, and stopping rule. When the set holds only the frozen model, U describes the model-alone system. As components are added to the set, the same notation carries through unchanged, which is the point of writing it this way.

The interaction contrast. $\Gamma(A; B)$ is the signed difference between the joint gain and the sum of isolated gains under one matched evaluation contract. Four scores identify this nonadditivity, not a mechanism or the necessity of either component. Chapter 2 defines it. ρ_{Γ} divides the contrast by the positive total joint gain. Nonnegative isolated gains and a nonnegative contrast additionally make the ratio a fraction between zero and one. Negative contrasts remain meaningful signed quantities.

Part I symbols

Symbol	Meaning	Introduced
$K_{\theta}(z \mid c)$	Model conditional law over outputs given context	1
$U_{\mathcal{V}}(S)$	Population task score of assembly S	1
$\hat{U}_{\mathcal{V}}(S)$	Finite-sample estimate of that task score	1
dec	Decoding map from a distribution to a candidate output	1
eval_{τ}	Evaluator with threshold, parser, or tolerance τ	1
θ	Model parameters, frozen unless a chapter says otherwise	1
$\mathbb{E}$	Expectation, as in Chapters 1 and 22 (Chapters 6, 7, 10, 11, 13, 15, 16, and 19 write the same operator as roman E; a plain subscript label such as Chapter 27's p_E is not this operator)	1
$\mathcal{V}$	Declared evaluation contract: task distribution, scoring rule and units, sampling settings, budget, and stopping rule	1
q, n	Per-token success probability and target length in the exact-match example, so the score is q^n	1
$\Gamma(A; B)$	Interaction contrast between component sets	2
ρ_{Γ}	Interaction share of total joint gain	2
q_m	Per-attempt success probability for family member m	3
$\text{pass}_{\tau}(q_m)$	Thresholded pass indicator on q_m	3
$L(C) = aC^b + c$	Fitted pretraining-loss curve; C is training compute and a, b, c are fitted constants	3
ψ	Fitted forecast parameters	3
f_{ψ}, ϵ_m	Fitted forecast function and its error for member m	3
Γ_m	Interaction contrast measured for family member m	3
z_m	Declared predictor features for member m	3
G	Action graph: vertices are controller states, edges are allowed transitions	4
V_0	Initial vertex set	4
step	One-transition successor map on vertex sets	4
$\text{Reach}_G(V_0)$	Reachable set	4

Symbol	Meaning	Introduced
$\text{Act}(v)$	Candidate actions at vertex v	4
$\mathcal{G}$	Permission grant rule: maps a set of candidate actions to the permitted subset	4
$\text{Allowed}_{\mathcal{G}}(v)$	Permitted actions after filtering	4
$\Pi_{G,v}(p)$	Infinite-cluster probability	4
$p_c(G, v)$	Critical probability	4
d	Branching factor or degree	4
$x_t = (c_t, m_t, w_t, b_t)$	Composite agent state realization	5
w_t	World or environment state	5
$P(x_{t+1} \mid x_t)$	Composite transition kernel	5
Mem	Memory map from the whole history to the retained memory m_t ; the update itself is part of Upd	5
$\pi_{\theta}(z, a \mid x)$	Proposal policy: model law times decoder	5
κ	Coarse-graining map	5
$H(X), I(X; Y)$	Shannon entropy, mutual information	5
$\text{Grant}(g \mid x, u)$	Normalized permission-outcome law, including denial	5
$\text{Env}(w', o \mid x, u, g)$	Joint next-world and observation law	5
$\text{Upd}(c', m', b' \mid x, u, g, w', o)$	Joint context, memory, and budget update law	5
$P_{\text{env}}(x' \mid x, u)$	Controlled transition given a full proposal	5
$P_{\theta}(x' \mid x)$	Closed-loop transition after averaging over proposal policy	5
$u = (z, a)$	Full proposal: model output and decoded action	5

Symbols introduced after Part I

Symbol	Meaning	Introduced
$u(y)$	Utility of a single outcome	6
y	An outcome of an action	6
$\mathcal{A}$	The action set available at a state	6
$EU(a)$	Expected utility of an action	6
CE	Certainty equivalent	6
Ret_t	Return, the discounted sum of future rewards	7
r_t	Reward received on the transition out of x_t	7
γ	Discount factor	7
$\pi(a \mid x)$	Stochastic policy	7
V^π, V^*	State value under a policy, and under an optimal policy	7
Q^π	Action value under a policy	7
$\Phi(x)$	Local shaping potential	7
$\mathcal{X}$	The finite state set	8
$\mathbf{b}_t$	Belief, a distribution over $\mathcal{X}$ at time t	8
$\mathcal{B}$	The belief simplex, the set of distributions over $\mathcal{X}$	8
$Obs(o \mid x, a)$	Observation kernel: the chance of seeing o on arriving in x after a	8
$\bar{r}(\mathbf{b}, a)$	Expected reward of action a under belief $\mathbf{b}$	8
$J(x', o \mid x, a)$	Filtered joint next-state and observation law	8
Γ_H, α	Finite-horizon policy-vector set and one vector	8
$VOI(O)$	Value of observing O before a declared decision; Chapter 27 writes $VOI(o)$ for one realized observation	8
$g(v), h(v)$	True cost from the start to vertex v , and true cost from v to the nearest goal	9
$f(v)$	Cost of the best path constrained to pass through v	9
$\hat{g}(v), \hat{h}(v), \hat{f}(v)$	The computable estimates of each	9

Symbol	Meaning	Introduced
$\text{cost}(u, v)$	Cost of the edge from u to v	9
$s, \mathcal{T}$	Start vertex and goal set	9
$\text{Exp}(A, G)$	Number of vertices algorithm A expands on graph G	9
$h(u, v)$	True minimum path cost from u to v , distinct from the heuristic estimate $\hat{h}(v)$	9
$\mathcal{I}_o$	Initiation set of option o : the states where the option may start	10
π_o	Option o 's internal policy, mapping execution information to a primitive action	10
β_o	Option o 's termination condition	10
μ	Policy over options: the parent policy that selects which option to run	10
a_t	Action selected at step t ; in Chapter 11, the next action selected after t completed pulls	11
$\text{Reg}(T)$	Regret accumulated over T rounds	11
$\hat{\mu}_a$	Empirical mean payoff of action a	11
$N_t(a)$	Number of pulls of action a among the t completed pulls, before the next decision	11
$\text{Gain}(O)$	Decision-relevant information in an observation	11
$\hat{V}_t(x)$	The agent's current estimate of the value of state x	12
δ_t	Temporal-difference error at step t	12
α	Step size for an update	12
λ	Weighting between the one-step estimate and the realized outcome	12
Ret_t^λ	The weighted return that interpolates between them	12
$\text{Pot}(x)$	Bounded shaping potential on declared state	13
r'_t, G'_0	Shaped reward and return; the original reward r and the return keep their meaning (Chapter 13 gives the return a local alias for Chapter 7's Ret_t)	13
ϕ	Trainable policy parameters, distinct from the frozen model parameters θ	13
$\hat{P}(x' x, a)$	Learned transition-model estimate	14

Symbol	Meaning	Introduced
$\widehat{\text{Obs}}$	Learned observation-kernel estimate	14
ϵ	Uniform one-step total-variation model error	14
τ_{samp}	Sampling temperature of a generative model	14
R	Upper bound on the common expected immediate state-action reward	14
H_{max}	Declared finite planning cap	14
R^*	Retained record set chosen under the token budget	15
$c(m), B, \lambda$ (Chapter 15)	Raw token cost of record m , the token budget, and decision-value units per token	15
$\epsilon, 2\epsilon$ (Chapter 15)	Summary value tolerance and the resulting regret bound	15
$\text{Cov}(k)$	Coverage at k : the chance that at least one of k samples is correct	16
$\text{Sel}(k)$	Selection at k : the chance that the returned sample is correct	16
$\hat{c}_{\text{success}}$	Total attempt cost divided by authorized confirmed completions	16
$\pi(k)$	Measured chance that the selector's top-ranked sample is correct given k candidates that contain at least one correct candidate	16
d	Probability that the selector scores one correct sample above one incorrect sample	16
$\text{eff}(T, x)$	State resulting from applying tool T in state x	17
$\text{resp}(T, x)$	Report returned by tool T in state x	17
$\text{undo}(T)$	Recovery tool for an effect of T	17
$\text{Rep}(x)$	Conservative set of eligible repeat attempts	17
$\text{Ready}(T, x)$	Current authorization and preconditions for a repeat attempt	17
$\text{Absent}(T, x)$	Authoritative terminal non-application of the original attempt	17
$\text{Restored}(T, x)$	Verified recovery of relevant consequences and preconditions	17
$\beta, c_{\text{dup}}, c_{\text{miss}}$	Belief that the effect already landed, cost of a duplicate effect, and cost of the effect never happening	17
$\text{Exec}_{\text{ui}}(\tilde{a} \mid x, a)$	Normalized realized-operation law	18
P_{ui}	Execution-composed world kernel	18

Symbol	Meaning	Introduced
Act_{ui}	Finite realized-operation set including declared denial/no-effect operations	18
$\mathcal{A}_{\text{rob}}(\mathbf{b})$	Intersection of correctly specified authorization sets over belief support	18
$\lambda_{\text{ui}}, \text{Fresh}(\Delta)$	Material-change rate and no-invalidating-change probability	18
$\mathcal{A}_{\text{auth}}(x)$	Commands whose protected effect is authorized in state x	18
$u_i(\pi_1, \pi_2)$	Payoff to party i under a pair of policies	19
$\pi_i^{(s)}$	Party i policy from training run s	19
$\text{diag}(M), \text{off}(M)$	Matched-run and cross-run mean returns	19
$\text{JPC}(M)$	Proportional cross-play loss	19
$\text{BR}_i(\pi_{-i})$	Best response by party i to the other policy	19
M	Matched matrix whose entry $M_{\{st\}}$ pairs run s against run t	19
$\text{Know}_i(F)$	Party i knows proposition F	20
$\text{MK}_{\mathcal{P}}^n(F)$	n levels of mutual knowledge among party set $\mathcal{P}$	20
$\text{CK}_{\mathcal{P}}(F)$	Common knowledge of F among party set $\mathcal{P}$	20
q_e	Nonnegative flow on traffic-network edge e	21
$\ell_e(q_e)$	Latency incurred on edge e at flow q_e	21
$\text{TL}(q)$	Total latency of a feasible network flow q	21
q^{NE}	Nash-equilibrium flow	21
q^*	Total-latency-minimizing flow	21
$\ell_e^{\text{mc}}(q_e)$	Marginal-cost latency imposed by edge e	21
$q_{\text{upper}}, q_{\text{lower}}, q_{\text{middle}}$	Flows on the three directed Braess routes S-U-T, S-L-T, S-U-L-T	21
γ_{cap}	Extra-traffic fraction in the bicriteria routing comparison	21
C	Declared episode-cost random variable for tail-risk accounting	22
$\text{CVaR}_{\alpha}(C)$	Conditional value at risk of C at declared tail level α	22
z	Candidate tail cutoff in CVaR's infimum representation	22
B, b_t	Declared bounded-horizon risk budget and remaining budget before decision t	22

Symbol	Meaning	Introduced
$\hat{r}_t(a_t)$	Conservative charged risk estimate for action a_t at time t	22
Ret^π	Expected discounted reward under policy π	22
$\mathcal{I}$	Set of states in which a named invariant holds	22
Cost_i^π	Expected discounted auxiliary-cost return	22
d_i	Declared limit on auxiliary cost i	22
$\mathcal{T}$	Declared threat model	22
ϵ_{safe}	Deliberate safety margin	22
$A_{C_i}^{\pi_k}(s, a)$	Constraint advantage for auxiliary cost i	22
ϵ_i	Largest candidate-policy expected constraint advantage	22
δ	Ideal CPO step-size bound	22
λ_{safe}	Fixed multiplier between reward and one auxiliary cost	22
$\mathcal{C}_{\text{parent}}, \mathcal{C}_{\text{child}}$	Declared capability sets of a delegating parent and a delegated child	23
M_t, e_t	Monitor's enforcement-layer record at time t , and one observed event	23
Mon	Monitor update map from prior record and one observed event to the next record	23
δ, v, r	Document identity, version, and recipient in the publish predicate	23
$\text{Approved}(M_t)$	Set of (δ, v, r) triples the monitor's current record authorizes for release	23
$\text{Publish}(\delta, v, r, M_t)$	Boolean predicate: is this exact document-version-recipient triple currently approved	23
$\mathcal{F}$	One declared, finite attack family under a fixed threat model	23
$\text{Risk}(\mathcal{F})$	Worst-case violation probability across $\mathcal{F}$	23
k	Number of samples drawn	24
p	Probability that one sample is correct	24
N_{eval}	Number of evaluated task instances	24
$\Delta_{\text{match}}(\theta)$	Score difference for frozen model θ between a named benchmark and its matched evaluation	24

Symbol	Meaning	Introduced
ζ	Declared threshold on a continuous metric when defining a thresholded scale onset	24
Onset_ζ	First declared family scale at which the thresholded metric reaches ζ	24
$\mathcal{K}$	Fixed same-bank task set used to compare coverage with deployed selection	24
$\widehat{\text{Cov}}_\mathcal{K}(k), \widehat{\text{Sel}}_\mathcal{K}(k)$	Empirical same-bank coverage and deployed-selector success after k candidates per task	24
$\mathcal{C}, \xi$	Finite tested configuration set and one configuration index	24
Pr	Probability over Equation 24.4's declared sampling or resampling design	24
$\text{Lower}_{\alpha_{\text{tail}}}^{\text{sim}}(\xi)$	Lower score bound for configuration ξ from a procedure with joint coverage across all tested configurations	24
SE_{match}	Estimated standard error of a difference of independent bank pass proportions	24
α_{tail}	Tail probability level for the simultaneous lower-bound guarantee	24
$\text{LCF}_{\alpha_{\text{tail}}}$	Simultaneous lower confidence frontier across tested configurations	24
ϕ_t, ϕ'	Parent and candidate versions of the agent procedure around frozen weights	25
$\widehat{\Delta}_G(\phi'), \mathbf{G}$	Guard-set estimate and its frozen guard cases for the named candidate	25
$\widehat{C}_G(\phi'), \mathcal{G}$	Guard cost estimate and authority envelope for the proposed release	25
$\tau, \mathbf{C}, \varepsilon_{\text{safe}}$	Local uplift threshold, cost limit, and reserved safety margin	25
$\mathbf{D}, \widehat{V}_D$	Development set and a procedure's estimated value on it	25
$\mathcal{M}, \Phi_t$	Declared bounded family of procedure changes and the finite candidate collection compared on $\mathbf{D}$	25
$\mathbf{m}, Z_i(\phi')$	Number of guard cases and the paired candidate-minus-parent outcome on guard case $\mathbf{i}$, bounded in $[-1, 1]$	25
$\Delta(\phi')$	Population mean of $Z_i(\phi')$: the candidate's true uplift, an unknown parameter	25
$\text{Authorized}_\mathcal{G}, \text{Rollback}$	Boolean authority predicate for the proposed release and the check for a tested path back to the parent	25
$\mathbf{n}, \mathbf{c}$	Single-task aliases of $\mathbf{n}_i, \mathbf{c}_i$, the source-pool size and the number of passing samples for task $\mathbf{i}$	26

Symbol	Meaning	Introduced
$\text{Del}(d, \Delta)$	Delegation action to destination d , with stated maximum delay Δ before a response, expiration, or fallback	27
$p_E(x)$	Probability that delegated destination E returns a correct resolution at input or state x	27
$\mathcal{R}_{\text{auth}}(x)$	Authorized decision routes in state x	27
ϱ	One member of $\mathcal{R}_{\text{auth}}(x)$, an authorized decision route	27
$V(\varrho, x)$	Declared value of authorized route ϱ in state x	27

Chapter 25 local notation

Chapter 25 uses $\phi_t, \phi', G, \hat{\Delta}_G, \hat{C}_G, \tau, c, \varepsilon_{\text{safe}}$, and $\mathcal{G}$ only for its constructed procedure-release rule. θ keeps its book-wide meaning: frozen model parameters. These local release symbols do not rename the book's standing state, action, score, or authority notation.

Notes on symbols in Chapters 13, 16, and 18

$\text{Pot}(x)$ in Chapter 13 is a bounded shaping potential; its terminal boundary affects policy comparisons. Chapter 16's cost per authorized confirmed success is undefined when no run succeeds. In Chapter 18, Exec_{ui} connects a proposed command to a realized operation. $\mathcal{A}_{\text{rob}}$ retains actions feasible throughout the declared belief support. $\text{Fresh}(\Delta)$ describes no material interface change under a stated constant-rate model.

Using the guide

Local indices, declared aliases, and diagram labels retain their stated scope; return to the defining chapter for domains and assumptions.